# MILITARY SPECIFICATION

## TORPEDO TUBE, SURFACE VESSEL, MARK 32 MOD 14

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification established the requirements for the procurement of Torpedo Tube Assembly Mk 32 Mod 14 for use on surface vessels.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-C-104 | Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted |
| MIL-P-116 | Preservation, Methods of |
| MIL-S-901 | Shock Tests, H.I. (High-Impact): Shipboard Machinery, Equipment and Systems, Requirements for |
| MIL-C-16173 | Corrosion Preventative Compound, Solvent Cutback, Cold-Application |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-H28 | Screw-Thread Standards for the Federal Service |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this documents should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524) Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-167 | Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II Internally Excited) |
| MIL-STD-810 | Environmental Test Methods |

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein:

Naval Ordnance Systems Command

| | |
|---|---|
| LD 632022 | Torpedo Tube, Surface Vessel, Mk 32 Mod 14 (Assembly), Index to List of Drawings |
| LD 632023 | Torpedo Tube, Surface Vessel, Mk 32 Mod 14 (Assembly) |
| 1623922 | Pressure Gage |
| 2320787 | "GO" Gage, Barrel (Assembly) Torpedo Tube, Surface Vessel, Mk 32 Mods 0 thru 14 |
| 2652289 | Dummy Torpedo Mk 46 Mod 1 |
| 3014043 | Torpedo Tube, Surface Vessel, Mk 32 Mod 14 (Assembly) |
| 3022829 | Torpedo Tube, Surface Vessel, Mk 32 Mod 14 External Wiring Diagram |
| 3022834 | Torpedo Tube, Surface Vessel, Mk 32 Mod 14 System Schematic |
| 3245120 | Pressure Fitting Adapter |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The Torpedo Tube Mark 32 Mod 14 shall be as specified herein and in conformance with LD 632023.



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3.2 Detailed requirements

3.2.1 First Article Sample. Unless otherwise specified, a sample shall be subject to first article inspection (see 4.2 and 6.2).

3.2.2 Materials. Unless otherwise specified in the contract, requisition or order, all materials and parts entering into the final assembly of Torpedo Tube Mk 32 Mod 14 shall conform to the detailed requirements given on the drawings and specifications listed in LD 632023 and subset LD's/PL's.

3.2.3 Finishes, coatings, constructions, and dimensions. Finishes, coatings, constructions, and dimensions shall be in accordance with the applicable drawings, specifications, and publications listed in LD 632023 and subset LD's/PL's.

3.2.4 Moisture, fungus, and corrosion. The tubes shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on drawings and specifications in LD 632023 and subset LD's/PL's.

3.2.5 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in LD 632023 and subset LD's/PL's.

3.2.6 Threads. Unless otherwise specified, all threads shall be in accordance with FED-STD-H28. The class or fit for threads shall be as specified on the applicable drawings.

3.3 Performance requirements and product characteristics. Torpedo Tube Mk 32 Mod 14 shall meet the following performance requirements and product characteristics:

3.3.1 Hydrostatic pressure (breech mechanisms). The breech mechanisms shall withstand a hydrostatic pressure of 4500 plus 20 minus 0 psig without failure of any parts (see 4.4.5.1).

3.3.2 High pressure pneumatic system pressure drop. With the securing mechanism removed, the pressure in the pneumatic system of each barrel shall stabilize at 1350 psig minimum. With the securing mechanism installed, the pressure shall stabilize at 750 psig minimum. There shall be no deformation or failure of any part of the pneumatic system (see 4.4.5.3).

3.3.3 Hydrostatic pressure, low pressure pneumatic system. The low pressure pneumatic system of each barrel shall withstand a hydrostatic pressure of 225 psig without failure of any part. (see 4.4.5.2)

3.3.4 Breech mechanism air loss. After being initially charged to a pressure of 2000 plus or minus 50 psig, allowed to cool for 2 hours, and then bled down to 1600 plus or minus 10 psig, none of the three breech mechanisms shall lose more than 5 percent air pressure, due to leakage, over a 24-hour period. (see 4.4.5.4)



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3.3.5 Bore size of barrel. Each barrel shall pass a "GO" gage (DWG 2320787) with an axial force not to exceed 150 pounds. (see 4.4.4.5.5)

3.3.6 Electrical requirements

3.3.6.1 Dielectric strength. Electrical circuitry of the tube shall withstand the application of a potential of 1000 volts ac between mutually insulated terminals and ground for one minute minimum without breakdown. (see 4.4.5.6)

3.3.6.2 Insulation resistance. Immediately following the dielectric strength test, the insulation resistance between hipotted points in the barrel circuits shall be 3 megohms minimum when a potential of 500 volts dc is applies. The insulation resistance between the hipotted points of the training gear heater circuit shall be 0.5 megohm minimum when a potential of 500 volts dc is applied. (see 4.4.5.7)

3.3.6.3 Circuitry. Electrical circuitry shall be in accordance with NAVSEA Dwg 3022834. (see 4.4.5.8)

3.3.6.4 Input/Output signals. The torpedo tube shall have the inputs and outputs listed in TABLES I and II. External wiring shall be in accordance with Dwg 3022829. (see 4.4.5.9 and 4.4.5.10)

3.3.7 Squib fire switches. The switch plunger shall move out forcefully when the latch place is released. (see 4.4.5.11)

3.3.8 Training gear crank test. When the base of the torpedo tube is level, the force required to initiate movement of the training gear, applied tangentially to the arc described by the training gear crank handle, shall be not greater than 30 pounds. (see 4.4.5.13)

3.3.9 Limit stops. The limit stops shall prevent an arc of train in excess of 190 degrees.

3.3.10 Muzzle cover seal test. The axial force required to remove the muzzle cover shall be 30 plus or minus 10 pounds. (see 4.4.5.14)

3.3.11 Control box ready light switch. When the switch actuator plate on each control box is placed in the ON position, the ready light switch contact shall close. When place in the OFF position, the ready light switch shall be open with no continuity reading indicated. (see 4.4.5.16)

3.3.12 Pressure switch operation. The pressure switch contacts in each barrel shall close at 1275 psig plus or minus 75 psig with increasing pressure and remain closed at pressures above this value. The pressure switch contacts shall open at 1275 plus or minus 75 psig on falling pressure and remain open at pressures below this value. (see 4.4.5.16 and 4.4.5.24)



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3.3.13 Breakaway plug puller mechanism

3.3.13.1 The lever of the breakaway plug puller mechanism shall be capable of being latched and unlatched by hand. (see 4.4.5.15 and 4.4.5.23)

3.3.13.2 The breakaway plug puller mechanism shall permit dry firings when in the latched position, shall remain in the latched position at the completion of these firings and shall be capable of being unlatched by hand. (see 4.4.5.17.3 and 4.4.5.23)

3.3.13.3 The breakaway plug, during torpedo firings, shall pull away from the torpedo receptacle in Torpedo Mk 46 without damage and shall not affect torpedo exit velocities when the tube is fired at pressures of 1200 through 1500 psig. (see 4.4.5.23)

3.3.14 Control value operating lever. Each barrel of the tube shall not be capable of being fired, either electrically or manually, when the control valve operating lever is in the STAND BY condition. There shall be no loss of pressure breech mechanism.

3.3.15 Component voltage. Electrical components for each barrel shall operate throughout an input voltage range as specified on applicable drawings.

3.3.16 Manual firing. The axial force that must be applied on the manual firing button to initiate the firing cycle at pressures of 1200 through 2000 psig shall not exceed 25 pounds. (see 4.4.5.17.2 and 4.4.5.22)

3.3.17 Dry firing. Each barrel of the torpedo tube assembly shall be capable of being fired, without a torpedo, by mechanical means at any flask pressure from 1200 to 2000 psig and by electrical means when the solenoid valves are energized for 400 milliseconds maximum at any flask pressure of 1350 through 2000 psig without damage, malfunctions, or hindrance to subsequent firings with a torpedo. (see 4.4.5.17)

3.3.17.1 During dry firings of the tube and firings of dummy Torpedo Mk 46 Mod 1 at pressure of 1200 through 1500 psig, the plug puller mechanism shall move automatically from the unlatched to the latched position. Upon completion of firings, the lever shall be capable of being unlatched by hand. (see 4.4.5.17.6)

3.3.17.2 At the the completion of the dry firing tests there shall be no damage sustained by the torpedo tube assembly. (see 4.4.5.17.7)

3.3.18 Breech mechanism interchangeability. Each breech mechanism assembly shall be capable of operating within every barrel of the tube assembly.

3.3.19 Ready-light effect on electrical firing capability. Each barrel of the torpedo tube assembly shall be capable of being electrically fired at 1350 psig minimum under conditions that will close all three switches in the ready light circuit (firing interlock, pressure, and ready light).



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TABLE I. Inputs to Starboard Torpedo Tube Mk 32 Mod 14

| Function | Signal Characteristics | | Input |
|---|---|---|---|
| | Voltage | Current (Max) | Connector Pin |
| Stow Mount Order Common | 115 vac<br>60 Hz<br>Ø B | 104 mA | J302-T |
| Make Ready 1, 3, and 5 | 115 vac<br>60 Hz<br>Ø A | 200 mA | J302-A<br>J302-E<br>J302-B |
| Make Ready Common | 115 vac<br>60 Hz<br>Ø B | 200 mA | J302-F |
| Firing Sector Clear Supply 26 VDC | 26 vdc | 6 mA | J302-G |
| Interlock Supply 26 VDC | 26 vdc | 155 mA | J302-D |
| Interlock Supply Common 26 VDCC | 26 vdcc | 155 mA | J302-H |
| BMCB In Remote Supply 26 VDCC | 26 vdcc | 6 mA | J302-K |
| Prepare Mount Order | 115 vac<br>60 Hz<br>Ø A | 104 mA | J302-L |
| Prepare Mount Order Common | 115 vac<br>60 Hz<br>Ø B | 104 mA | J302-S |
| Stow Mount Order | 115 vac<br>60 Hz<br>Ø A | 104 mA | J302-M |
| Darken Ship Supply 26 VDC | 26 vdc | 120 mA | J302-W |
| Tube Loaded | 26 vdcc | 5 mA | J301-Q |
| Continuity Monitor | 26 vdcc | 5 mA | J301-Z |



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TABLE I. Inputs to Starboard Torpedo Tube Mk 32 Mod 14 (continued)

| Function | Signal Characteristics | | Input |
|---|---|---|---|
| | Voltage | Current (Max) | Connector Pin |
| Gyro Sense | 0 - 26 vdc | 30 mA | J301-R |
| Search Depth Sense | 0 - 26 vdc | 30 mA | J301-W |
| Tube Ready Supply 26 VDC | 26 vdc | 5 mA | TB1-7 |
| Torpedo Reference Common 26 VDCC | 26 vdcc | 3 A | TB1-19 |
| Torpedo Reference Sense Common | 26 vdcc | 60 mA | TB1-19 |
| Gyro Drive 1, 3, and 5 | 26 vdc | 2.75 A | TB1-13<br>TB2-13<br>TB3-13 |
| Search Depth Drive 1, 3, and 5 | 26 vdc | 2.75 A | TB1-16<br>TB2-16<br>TB3-16 |
| Torpedo Reference 1, 3, and 5 26 VDC | 26 vdc | 3 A | TB1-20<br>TB2-20<br>TB3-20 |
| Squib Fire 1, 3, and 5 | 26 vdc | 2 A | TB1-4<br>TB2-4<br>TB3-4 |
| Fire Order Tube 1, 3, and 5 | 115 vac<br>60 Hz<br>Ø C | 2 A | TB1-11<br>TB2-11<br>TB3-11 |
| Fire Order Common Tube 1, 3, and 5 | 115 vac<br>60 Hz<br>Ø B | 2 A | TB1-10<br>TB2-10<br>TB3-10 |
| Door Open | 115 Vac<br>60 Hz<br>Ø B | --- | J303-A |
| Ship's Lighting Supply 115 VAC | 115 vac<br>60 Hz<br>Ø B | --- | J303-G |
| Ship's Lighting Supply Common 115 VACC | 115 vacc<br>60 Hz<br>Ø A | --- | J303-J |



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TABLE I. Inputs to Starboard Torpedo Tube Mk 32 Mod 14 (continued)

| Function | Signal Characteristics | | Input |
|---|---|---|---|
| | Voltage | Current (Max) | Connector Pin |
| Train Motor Power | 115 vac<br>60 Hz<br>Ø A | 4 A | J202-C |
| Train Motor Power | 115 vac<br>60 Hz<br>Ø B | 4 A | J202-E |
| Train Motor Power | 115 vac<br>60 Hz<br>Ø C | 4 A | J202-D |
| BMCB Power | 115 vac<br>60 Hz<br>Ø A | 4 A | J202-H |
| BMCB Power | 115 vac<br>60 Hz<br>Ø B | 4 A | J202-A |
| Emergency Squib Fire 35 VAC | 35 vac<br>60 Hz | 2 A | TB105-7 |
| Emergency Squib Fire 35 VACC | 35 vacc<br>60 Hz | 2 A | TB102-8 |



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TABLE II. Outputs from Starboard Torpedo Tube Mk 32 Mod 14

| Function | Signal Characteristics | | Output |
|---|---|---|---|
| | Voltage | Current (Max) | Connector Pin |
| Tube Ready 1, 3, and 5 | 26 vdc | --- | TB1-6<br>TB2-6<br>TB3-6 |
| Gyro Sense 1, 3, and 5 | 0 - 26 vdc | --- | TB1-15<br>TB2-15<br>TB3-15 |
| Search Depth Sense 1, 3, and 5 | 0 - 26 vdc | --- | TB1-18<br>TB2-18<br>TB3-18 |
| Tube Loaded 1, 3, and 5 | 26 vdc | -- | TB1-14<br>TB2-14<br>TB3-14 |
| Continuity Monitor 1, 3, and 5 | 26 vdcc | --- | TB1-8<br>TB2-8<br>TB3-8 |
| Open Door | 115 vac<br>60 Hz<br>Ø B | --- | J303-D |
| Close Door | 115 vac<br>60 Hz<br>Ø B | --- | J303-B |
| Door Common | 115 vac<br>60 Hz<br>Ø A | --- | J303-C |
| Firing Sector Clear | 26 vdc | --- | J302-C |
| BMCB In Remote | 26 vdc | --- | J302-R |
| Darken Ship | 26 vdc | --- | J302-X |



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TABLE II. Outputs from Starboard Torpedo Tube Mk 32 Mod 14 (continued)

| Function | Signal Characteristics | | Output |
|---|---|---|---|
| | Voltage | Current (Max) | Connector Pin |
| Torpedo Reference Common 26 VDCC | 26 vdcc | 3 A | J301-A |
| Gyro Drive | 26 vdc | 2.75 A | J301-E |
| Search Depth Drive | 26 vdc | 2.75 A | J301-K |
| Squib Fire | 26 vdc/ 35 vac | 2 A | J301-Y |
| Torpedo Reference 26 VDC | 26 vdc | 3 A | J301-N |
| Fire Order Tube 1, 3, and 5 | 115 vac 60 Hz Ø C | 2 A | J301-C J301-D J301-J |
| Fire Order Common Tube 1, 3, and 5 | 115 vac 60 Hz Ø B | 2 A | J301-G J301-H J301-P |
| Torpedo Reference Sense Common | 26 vdcc | 60 mA | J301-L |



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3.3.20 Torpedo exit velocities. The minimum exit velocities of dummy Torpedo Mk 46 Mod 1 (508 plus or minus 8 pounds) for each barrel of the torpedo tube assembly shall be 40 feet per second for each test. (see 4.4.5.20)

3.3.21 Electrical continuity. Electrical continuity of Torpedo Tube Assembly Mk 32 Mod 14 shall be in accordance with NAVSEA Dwg 3022834 and 4.4.5.18.

3.3.22 Lanyard retainers. The spring-loaded retainers shall be capable of being depressed easily by hand and shall be capable of accepting and securing a lanyard wire of 0.069 minimum diameter. (see 4.4.5.19)

3.3.23 Bulkhead mounted control box. The Bulkhead Mounted Control Box shall be capable of selecting either REMOTE or LOCAL operation of the torpedo tube and firing door.

3.4 Environmental requirements

3.4.1 Vibration

3.4.1.1 Vibration, nonoperating. The torpedo tube assembly, less packing or packaging, shall be capable of meeting all quality conformance tests after being subjected to vibration test of MIL-STD-167 for Type I vibrations in the three mutually perpendicular axes. (4.5.1.1)

3.4.1.2 Vibration, standby condition. The torpedo tube assembly, loaded with three dummy torpedoes and containing breech assemblies pressurized to 1500 psig, shall be capable of meeting all quality conformance tests during and after being subjected to the vibration test of MIL-STD-167 for Type I vibrations in the three mutually perpendicular axes. (4.5.1.2)

3.4.2 Shock

3.4.2.1 Shock nonoperating. The torpedo tube assembly, less packing or packaging, shall be capable of meeting all quality conformance tests after being subjected to the shock impulses of the magnitude and number specified in MIL-S-901 for medium weight equipment. (4.5.2.1)

3.4.2.2 Shock, loaded condition. The torpedo tube, loaded with three dummy torpedoes, shall be capable of meeting all quality conformance tests after being subjected to the shock impulses of the magnitude and number specified in MIL-S-901 for medium weight equipment. (4.5.2.2)

3.4.3 Temperature and humidity

3.4.3.1 High temperature operation. When conditioned and tested in accordance with 4.5.3.1, the torpedo tube shall meet all the performance requirements and product characteristics of this specification and shall show no corrosion or damage.



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3.5 Cleanup. Prior to and after final assembly, all parts and components shall be thoroughly cleaned of all loose, spattered, or excess solder, metal chips, and other foreign material. All burrs and sharp edges as well as flashing shall be removed.

3.6 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified in the contract or order (see 6.2), the Government reserves the right to conduct any or all of the tests and examinations of this specification after installation of the torpedo tube aboard a Naval vessel.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3 and 4.5)
b. Periodic production inspection (see 4.3 and 4.5)
c. Quality conformance inspection (see 4.3 and 4.4)

4.3 Sampling

4.3.1 First article sample. A first article sample of one torpedo tube assembly, manufactured in accordance with 3.2.1 and preserved, packaged and packed as specified in the contract or order, shall be subjected to the tests detailed in 4.5, at an activity designated by the procuring activity. (see 6.2)

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.



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4.3.3 Periodic production sample. From each succeeding lot of 50 or less, a periodic production sample of one torpedo tube assembly shall be selected at random and submitted to an activity designated by the procuring activity (see 6.2) for periodic production inspection. The sample shall be examined for compliance with the requirements of the contract, specification and drawings and tested in accordance with 4.5. Failure of the sample to comply with any requirements shall result in the rejection of the lot as determined by the procuring activity. (see 6.2)

4.3.4 Quality conformance sampling. The tests and examinations detailed in 4.4 should be conducted on each torpedo tube assembly to be delivered; therefore there shall be no sampling for quality conformance. Defective units shall be rejected.

4.4 Quality conformance inspection. Each torpedo tube Mk 32 Mod 14 to be delivered under contract shall be subjected to the tests and examinations of this section.

4.4.1 Test conditions. Unless otherwise specified, the torpedo tubes shall be sujected to quality conformance tests under the following conditions:

4.4.1.1 Test sequence. The sequence of testing procedures presented herein is suggested only. The tests may be performed in any order that will permit the most practical and economical use of manpower and equipment.

4.4.1.2 Temperature. Room ambient 65 degrees Fahrenheit (F) to 95 degrees F (18 degrees centigrade (C) to 35 degrees C).

4.4.1.3 Altitude. Normal ground atmospheric pressure.

4.4.1.4 Vibration. None

4.4.1.5 Humidity. Room ambient to 95 percent relative maximum.

4.4.2 Test results. When specified in the contract or order (see 6.2), test results shall be recorded, or verification noted, as required, opposite corresponding paragraph numbers on the Quality Conformance Inspection Data Sheet (Appendix) for each torpedo tube assembly inspected (see 5.4).

4.4.3 Acceptance criteria. Failure of Torpedo Tube Mk 32 Mod 14 to pass any examination or test of this section shall be cause for rejection.

4.4.4 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance test. To perform the tests, the following items of test equipment, or equivalent, are required:

4.4.4.1 A means for producing a hydrostatic pressure of 4500 psig.



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4.4.4.2 A water pressure gage capable of reading 4500 psig with an accuracy of plus or minus ¼ percent.

4.4.4.3 A 0-3000 psig high pressure air source, equipped with an air line filter, to remove water, oil, and dirt, which shall not pass particles larger than 50 microns.

4.4.4.4 An air pressure gage of 0 to at least 3000 psig with an accuracy of plus or minus ½ percent at 2500 psig.

4.4.4.5 A "Go" gage for the barrel in accordance with Dwg 2320787.

4.4.4.6 Standard high potential test equipment capable of an output of 1000 volts, 60 to 100 Hz.

4.4.4.7 Megohm bridge, 500 volts dc, with a range of 0.1 to 1,000,000 megohms.

4.4.4.8 Multimeter, Simpson model 260 or equivalent.

4.4.4.9 A power source capable of supplying: (a) 115 plus or minus 0.5 volts ac, 60 Hz, single phase; the voltage must be capable of being varied from at least 102 to 124 volts ac, (b) 115 Vac 60 Hz 3 phase and (c) 26 to 28 Vdc.

4.4.4.10 Test cabinet/power supply (continuity/operational tests).

4.4.4.11 An ac voltmeter capable of measuring 102 to 124 volts ac with an accuracy of at least plus or minus 1.125 volts.

4.4.4.12 A means for measuring the angle of rotation of the training gear to the nearest ½ degree.

4.4.4.13 A pull scale with a memory pointer and a capacity of at least 155 pounds, with a maximum error of plus or minus 10 percent of the reading.

4.4.4.14 A test circuit capable of supplying 102 to 124 volts, 60 Hz ac from the power supply of 4.4.4.10 to the solenoid for the period of 400 milliseconds maximum. Current capacity shall be at least 2 amperes. Can be included in the test cabinet (4.4.4.10).

4.4.4.15 Dummy Torpedo Mk 46 Mod 1 (Dwg 2652289) (weight: 508 plus or minus 8 pounds).

4.4.416 Launching ramp or track from which dummy torpedoes can be launched. The ramp or track shall include a means for decelerating and stopping the dummy torpedo on the track without damage.

4.4.4.17 Two infrared photoelectric control units suitable for use with electronic counter of 4.4.4.18.

4.4.4.18 An electronic counter capable of recording up to 10,000 milliseconds in one millisecond intervals with instant reset capabilities.

4.4.4.19 A continuity indicator.



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4.4.4.20 A 125 plus or minus 2 vdc power supply with a 2.7 plus or minus 0.1 k ohm series resistor and can be included in test cabinet (4.4.4.10).

4.4.4.21 Pressure fitting adapter (Dwg 3245120).

4.4.4.22 Stop watch.

4.4.4.23 Nitrogen bottle of sufficient volume to purge the breech mechanism air flasks.

4.4.4.24 Sound level meter/octave analyzer.

4.4.4.25 A 0.024 plus or minus 0.0005 inch diameter orifice for the high pressure air source of 4.4.4.3.

4.4.4.26 A means to lift the torpedo tube to gain access to the training gear cavity.

4.4.5 Test procedures. The following tests shall be performed:

WARNING

When working with high pressure air or electrical apparatus, observe all safety precautions. While conducting any of the firings, personnel shall stand clear of the breechward access opening and breech mechanism. Charging lines should be secured with safety chains when high pressure air charging operations are in progress.

4.4.5.1 Hydrostatic pressure test (breech mechanism). With the breech mechanisms removed from the tube assembly, remove pressure gage, (1623922), and insert a standard 1/8-pipe plus in the void. The breech mechanisms shall be filled with water and pressurized to 4500 plus 20 minus 0 psig. This pressure shall be maintained for 10 minutes minimum without failure of any part. At the completion of this test reduce the pressure to zero, drain the breech mechanism and purge with dry nitrogen. Reinstall the pressure gage.

4.4.5.2 Hydrostatic pressure test of low pressure pneumatic system. Fill the 100-150 psig system with water and pressurize to 225 psig. Maintain this pressure for 10 minutes with the barrel in the safe condition and for 10 minutes with the barrel in the ready condition. Perform this test on all barrels. After completion of the hydrostatic test, thoroughly purge water from the system with nitrogen.



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4.4.5.3 High pressure pneumatic system pressure drop test. This test shall be performed on each barrel of the torpedo tube assembly. Remove the breech mechanism and securing mechanism from the barrel. Secure the nipple attached to the manifold and piping assembly to a shutoff valve or other suitable blanking device. Adjust the air source to 2000 plus 0 minus 20 psig and attach it through a 0.024 plus or minus 0.0005 inch diameter orifice to the barrel pneumatic system using a pressure fitting adapter (Dwg 3245120). Actutate the firing solenoid valve and allow the pneumatic system line pressure to stabilize. Measure the pressure on the barrel side of the orifice. Repeat this test with the securing mechanism installed in the barrel. Measure the pressure.

4.4.5.4 Breech mechanism air loss test. With the breech mechanisms removed gradually charge each (using the test equipment of 4.4.4.3, 4.4.4.4, and 4.4.4.5) at a rate not exceeding 250 psig per minute to a pressure of 2000 psig pneumatic and permit them to cool for a period of 2 hours. The breech mechanisms shall then be bled to a pressure of 1600 plus or minus 10 psig. After a 24-hour period, observe the pressure reading, using the gage of 4.4.4.5. Compensation may be made for change in pressure due to temperature.

4.4.5.5 Bore size of barrel test. Perform the test using the "Go" bore gage (4.4.4.5). Measure the axial force required to go through each barrel.

4.4.5.6 Dielectric strength

4.4.5.6.1 Dielectric strength bulkhead mounted control box assembly. This assembly shall be subjected to a potential of 1000 Vac, 60 Hz applied between ground (terminal TB 304-12 in box) and all input terminals as shown on external wiring diagram 3022829 (schematic diagram 3022834) for one minute.

4.4.5.6.2 Dielectric strength torpedo tube assembly. This assembly shall be subjected to a potential of 1000 VAC 60 Hz applied between ground (terminal TB102-14 of Junction Box 5185872) and all input terminals as shown on external wiring diagram 3022829 (schematic diagram 3022834) for one minute.

4.4.5.7 Insulation resistance tests. For the purpose of the test the bulkhead mounted control box assembly and torpedo tube assembly shall be in the same condition as required for the dielectric strength tests. Immediately after the hipot tests, subject the insulation between the hipotted points to 500 Vdc and, the insulation between terminals TB 201 1, 2, 3 and 4 (mount barrel junction box) and ground to 500 VDC.



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4.4.5.8 Ground circuit. Continuity shall exist between TB102-14 of junction box 5185872 and pin G of teat receptacle J2 in the junction and torpedo tube junction boxes (5185872 and 5206053) and control boxes (3). In mount barrel junction box continuity shall exist between TB201-16 and the box. In bulkhead mounted control box continuity shall exist between TB304-12 and the box.

4.4.5.9 Firing solenoid operating current. Apply 115 plus or minus 0.5 volts ac accross terminals 10 and 11 of TB1; measure and record the current flow for barrel number 1. Repeat for barrel number 2 (TB2) and barrel number 3 (TB3). Total drawn by each solenoid should not exceed 1.5 amp overall (continuous).

4.4.5.10 Safe/Ready solenoid operating current. Apply 115 plus or minus 0.5 volts ac across terminals TB 201-1 and -4 (Barrel No. 1), TB 102-2 and -4 (Barrel No. 2) and between TB 201-3 and -4 in mount barrel junction box. Measure and record the current flow for each barrel. The total current drawn by each solenoid should not exceed 0.2 ampere max (continuous).

4.4.5.11 Squib fire switch. With the control box latch plates in the locked position, continuity shall exist across terminals 8 and 9 of TB101 and no continuity shall exist across terminals 7 and 8 and terminals 7 and 9 of TB101 in the control box of each barrel. Release the latch on the control box, observing that the switch plunger moves out forcefully, and verify that continuity exists across terminals 7 and 8 of TB101 in each barrel and than no continuity exists across terminals 8 and 9, and 7 and 9 of TB101.

4.4.5.12 Interlock switches S202 and 2203. The torpedo tube assembly shall be trained through an arc of 45 degrees from the zero degree relative position. In Mount Barrel Junction Box the continuity of the switch circuit shall be measured at TB201-7 and-8. The circuit shall be open at 0 degrees plus or minus 2.5 degrees. The circuit shall close at 45 plus or minus 2.5 degrees.

4.4.5.13 Training crank test. With the base of the torpedo tube assembly level, attach a pull scale to the training crank handle and pull with a steady motion tangential to the arc of the crank travel. Gear motor select lever must be in the manual position.



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4.4.5.14 Muzzle cover test. With the muzzle cover installed, attach a pull scale to the center of the cover handle and pull in a direction perpendicular to the plane of the cover.

4.4.5.15 Breakaway plug puller mechanism. Verify that the lever of the breakaway plug puller mechanism in each barrel can be latched and unlatched by hand without excessive force.

4.4.5.16 Pressure and ready light switch operation. In Junction Box and Frame Assembly (5206053) verify continuity between TB1-7 and TB 1-6 of barrel No. 1, TB1-7 and TB2-6 of barrel No. 2, and TB1-7 and TB3-6 of barrel No. 3.

4.4.5.16.1 With the ready light switch actuator plate rotated to the ON position and utilizing suitable means to indicate electrical continuity, gradually apply pressure to the air flask at a rate not exceeding 250 psig per minutes until the pressure switch closes.

4.4.5.16.2 Gradually decrease the pressure untill the pressure switch opens.

4.4.5.16.3 Ready light switch. Place the ready light switch in the ON position and note position of the contacts. With the pressure switch closed as in 4.4.5.16.1, place the ready light switch actuator plate in the OFF position. Note continuity reading. This test shall be performed on each barrel.

4.4.5.17 Dry firing tests. Each breech mechanism shall be charged gradually at a rate not exceeding 250 psig per minute to pressures in accordance with Schedule I. With the control valve operating level in the READY position, the plug puller mechanism in the unlatched position, the breakaway plug disconnected from the plug puller mechanism and secured to the guide, and all firing interlocks satisfied, each barrel shall be fired a total of six times without a torpedo in place. Four of the firings shall be made by electrical means and two by mechanical means. The power supply (4.4.4.10) shall be varied as specified in Schedule I for the four electrical firings. The application of voltage shall be regulated by the test circuit of 4.4.4.14, which shall limit firing potential duration to 400 milliseconds maximum. After each firing the breech mechanism shall be rotated to ensure that all three breech mechanisms will function in any of the three barrels.



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SCHEDULE I

Dry firing tests

| Firing Sequence | Flask pressure | AC input voltage to solenoid | Pressure Switch condition | Firing method |
|---|---|---|---|---|
| 1 | 1200 psig | - | - | Manual |
| 2 | 2000 psig | - | - | Manual |
| 3 | 1350 psig | 102.5 ± 0.5 | Closed | Electrical |
| 4 | 2000 psig | 102.5 ± 0.5 | Closed | Electrical |
| 5 | 1350 psig | 123.5 ± 0.5 | Closed | Electrical |
| 6 | 2000 psig | 123.5 ± 0.5 | Closed | Electrical |

NOTE: Prior to conducting the above tests remove the clevis pin from the lever of the plug puller mechanism and disconnect the lanyard from the lever. Replace and secure the clevis pin through the keeper and lever. Secure the breadaway plug to the guide by looping the lanyard over the plug.

4.4.5.17.1 Electrical firing test. Each barrel electrically fired shall fire with only one 400-millisecond application of test potential. Two firings shall be made at 102.5 plus or minus 0.5 volts ac and two at 123.5 plus or minus 0.5 volts ac.

4.4.5.17.2 Manual firing. During the manual firing tests measure the force required to activate the solenoid valve with the push scale.

4.4.5.17.3 Breakaway plug mechanism. After each dry firing, the plug puller mechanism shall be in the latched position. Verify that it can be unlatched by hand.

4.4.5.17.4 Control valve operating lever. During all dry firing tests observe that there is no leakage or premature air escape while the control valve operating lever is in the STANDBY condition. While performing the tests of Schedule I, with the control valve operating lever in the STANDBY condition, attempt to manually and electrically fire the tube. The tube shall not fire and there shall be no loss of pressure in the breech mechanism.



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4.4.5.17.5 Pressure switch operation. During the dry firing tests verify that the pressure switch condition is in accordance with Schedule I when the control valve operating lever is in the READY condition.

4.4.5.17.6 Breakaway plug mechanism. At the completion of the dry firing, check that the breakaway mechanism is in the latched position.

4.4.5.17.7 Observation for damage. At the completion of the dry firing tests inspect the torpedo tube assembly for damage.

4.4.5.18 Continuity. Check continuity in accordance with the applicable schematic diagram. Specific equipment and system continuity and inspection circuit tests are listed in 30.2.18, including BMCB interface.

4.4.5.19 Lanyard retainer (breechward access opening location). Depress the lanyard retainer plunger by hand and insert and assemble a loop of 0.069 inch minimum diameter wire.

4.4.5.20 Torpedo exit velocity. The two photoelectric control units (4.4.4.17) shall be spaced three feet apart alongside the track or ramp. The first set of cells shall be located the length of the torpedo ±0.5 foot from the muzzle end of the barrel. The photoelectric cells shall be attached to the electronic counter (4.4.4.18) so that when the first beam is broken by the nose of the torpedo the counter starts and when the second beam is broken by the nose of the torpedo, the counter stops. The average velocity shall then be calculated by the formula

$$V = \frac{d}{t}$$

Where d = distance in feet, V = average velocity, and t = time in seconds. Each breach mechanism shall be charge at the rate not exceeding 250 psig per minute. The breech mechanism shall be charged to a pressure of 200 psi greater than those pressures specified in Schedule II and allowed to cool for at least 5 minutes prior to being bled and readied for a firing in accordance with Schedule II.

Schedule II

Torpedo exit velocity tests

| Launchings per barrel (No.) | Torpedo type (Mk) | (Mod) | Air flask pressure (psig) | Pressure switch Condition | Firing Method |
|---|---|---|---|---|---|
| 1 | 46 | 1 | 1200 | - | Manual |
| 1 | 46 | 1 | 1350 | Closed | Electrical |
| 1 | 46 | 1 | 1500 | Closed | Electrical |



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4.4.5.21 Electrical firing test. Each electrical firing test shall be made with only one 400-millisecond application of test potential. The test potentail used for these tests shall be 115 plus or minus 0.5 volts ac.

4.4.5.22 Manual firing. During the manual firing tests measure with the push scale the force required to actuate the solenoid valve.

4.4.5.23 Breakaway plug mechanism

4.4.5.23.1 Prior to each firing the breakaway plug mechanism lever shall be placed in the unlatched position and the breakaway plug attached to the torpedo receptable.

4.4.5.23.2 After each firing the plug puller mechanism lever shall be in the latched position and shall be capable of being unlatched by hand.

4.4.5.23.3 After each firing the breakaway plug shall be inspected for damage.

4.4.5.24 Pressure switch operation. During the firing tests verify that continuity of the pressure switch circuit exists in accordance with Schedule II when the ready light switch is in the ON position and the control valve operating lever is in the READY CONDITION.

4.4.5.25 Control valve operating lever. During all tests observe that when the control valve operating lever is in the STANDBY condition there is no leakage or premature air escape.

4.4.5.26 Observation for damage. At the completion of the firing tests there shall be no damage to the barrel assemblies.

4.4.5.27 Packaging packing, and marking. The packaging, packing, and marking of the torpedo tube assembly shall be examined to ensure that it complies fully with the requirements of Section 5.

4.5 First article and periodic production tests. Each first article and periodic production sample, after satisfactorily passing the quality conformance tests and examinations (4.4), shall be subjected to the following environmental tests. Unless otherwise specified, the sample shall be deemed defective if it fails to satisfactorily pass the quality conformance tests and examination subsequent to these environmental tests.

4.5.1 Vibration tests

4.5.1.1 Vibration, nonoperation. The torpedo tube, secured to the mounting bracket of the testing machine in the same manner that it will be securing on shipboard, shall be subjected to the exploratory vibration test, varible frequency test, and endurance test as specified in MIL-STD-167 for Type I vibrations.



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4.5.1.2 Vibration, standby condition. A dummy Mk 46 Mod 1 Torpedo shall be loaded in each barrel. All three breech assemblies shall be pressurized to 1500 psig. The torpedo tube, loaded as specified and secured to the mounting bracket of the testing machine in the same manner that it will be secured on shipboard, shall be subjected to the exploratory vibration tests, variable frequency test, and endurance test as specified in MIL-STD-167 for Type I vibrations.

4.5.2 Shock tests

4.5.2.1 Shock, nonoperating. The torpedo tube assembly shall be attached to the anvil table of the shock testing machine by means of a standard mounting platform and a 30 degree mounting fixture as specified in MIL-S-901 for medium-weight equipment. The torpedo tube shall be oriented on the 30 degree mounting fixture such that the barrel conterlines are parallel to the horizontal plane. In each mounting condition the torpedo tube shall be subjected to Type A shock tests as specified in MIL-S-901 for medium-weight equipment.

4.5.2.2 Shock, loaded condition. The torpedo tube assembly shall be attached to the anvil table of the shock testing machine by means of a standard mounting platform and a 30 degree mounting fixture as specified in MIL-S-901 for medium-weight equipment. A dummy Mk 46 Mod 1 Torpedo shall be loaded in all barrels. The torpedo tube shall be oriented on the 30 degree mounting fixture such that the barrel centerlines are 30 degrees from the horizontal plane. In each mounting condition the torpedo tube shall be subjected to Type A shock tests as specified in MIL-S-901 for medium-weight equipment.

4.5.3 Temperature and humidity tests

4.5.3.1 High temperature and humidity tests. The torpedo tube assembly with fully pressurized breech assemblies (2000 psig), shall be placed within a test chanber and subjected to temperature and humidity cycling as specified in MIL-STD-810, Method 507, Procedure IV with the following exceptions:

(a) Referenced measurements of steps 4, 5, 6, and 8 may be deleted.

(b) The 24- hour cycles of step 5 shall have a high temperature minimum of 59°C (138°F) and a maximum as specified.

(c) The torpedo tube assembly shall be performance checked during the high temperature portion of the fifth cycle of step 5 prior to decreasing to 30°C (86°F).



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4.5.4 During and after performance of the above tests, no damage shall occur which chould affect proper operation or function of the tube.

4.5.5 At the completion of the above tests. The torpedo tube shall meet all the quality conformance tests of 4.4.

5. PREPARATION OF DELIVERY

.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning. Unless otherwise specified, the torpedo tube, in the nonoperating condition, shall be cleaned in accordance with Method C-1 of MIL-P-116.

5.1.1.2 Drying. Unless otherwise specified, the torpedo tube shall be dried in accordance with one or more of the procedures of MIL-P-116, as appropriate.

5.1.1.3 Preservation. Immediately after cleaning and drying the torpedo tube, the external unpainted metal surfaces which are subject to corrosion shall be preserved by the application of preservative compound conforming to the requirements of MIL-C-16173, Grade 2.

5.1.1.4 Unit packaging. Not applicable.

5.1.2 Level B. Non applicable.

5.1.3 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with the supplier's commercial practice when such practices have been approved by the procuring activity.

5.2 Packing

5.2.1 Level A



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5.2.1.1 Exterior containers. The torpedo tube assembly, in the non-operating condition, shall be packed in a nailed wood, skid-mounted crate with rubbing strips in accordance with MIL-C-104 for Class I, Style A, Type I crates. The base of the torpedo tube shall be rigidly fastened to the load-bearing members of the skid base and the tube barrels shall be braced and cushioned to prevent movement.

5.2.2 Level B. Not applicable.

5.2.3 Level C The torpedo tube assembly, in the nonoperating condition, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Special markings. None, unless otherwise specified (see 6.2)

5.3.2 Normal markings. In addition to the markings required by contract or order, shipping containers shall be marked in accordance with MIL-STD-129, including serial numbers.

5.4 Inspection data sheets. When Quality Conformance Inspection Data sheets are specified in the contract or order (see 6.2.1 and 6.2.2) for completion in accordance with 4.4.2, one data sheet (or copy thereof), bearing recorded data, shall be shipped with each torpedo tube assembly inspected. The torpedo tube assembly equipment serial number shall correspond to the serial number recorded on the accomplanying data sheet.

6. NOTES

6.1 Intended use. Torpedo Tube Assembly Mk 32 Mod 14 is used on Naval surface vessels to launch Torpedo Mk 46. Torpedo Tube Assembly Mk 32 Mod 14 is intended for use with Fire Control System Mk 116.

6.2 Ordering data. Procurements documents should specify the following:

6.2.1 Procurement requirements



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(a) Title, number, and date of this specification.

(b) When a first article sample is not required (see 3.2.1).

(c) When the Government waives the right to post-delivery inspection (see 4.1).

(d) Designated activity for first article and/or periodic production inspection(s) (see 4.3.1 and 4.3.3).

(e) Required level of preservation and packaging (see 5.1).

(f) Required level of packing (see 5.2).

(g) Whether any special markings are required (see 5.3.1).

(h) Requirement for submitting Quality Conformance Inspection Data Sheets (see 5.4).

(i) Any exceptions to this specification or its applicable drawings and documents.

(j) That all additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

(k) That any defects due to the contractor failure to comply with the drawings or specifications for this equipment, discovered during or after tests or examinations aboard a Naval vessel, shall be corrected by the contractor at no expense to the Government. provided the contractor is notified of such defects within the guarantee period of the contract.

(l) That production of torpedo tubes by the contractor prior to approval of the first article sample shall be at the contractor's risk. Approved first article samples shall be included as a part of the quantity specified for delivery in the contract or order.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirments List (CDRL). the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following:



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| Paragraph No. | Data requirements | Applicable DID no. | Option |
|---|---|---|---|
| 4.4.2 | Reports, Test, Specification Appendant | UDI-E-20602 | ----- |

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in MIL-P-116.

6.3 Definitions. For the purpose of this specification the following definitions are applicable:

6.3.1 Torpedo tube assembly. The launching device in its assembled state and compatible with starboard installation.

6.3.2 Nonoperating condition. The condition of the tube when assembled, prepared, issued, transported, and handled.

6.3.3 Loaded condition. That condition when a torpedo or dummy is installed in the tube barrel, secured in place by the securing mechanism, and the breech mechanism and muzzle covers are installed.

6.3.4 Safe condition. The condition of the barrel assembly when the lever of the control valve operating mechanism is in the muzzleward position, the latch over the manual fire button of the solenoid valve is up and locked in position. The muzzle covers installed, the Ready Light ON-OFF actuator in the OFF (down) position, no H.P. air in air flask (Breech Mechanism), tube trained to 0° firing angle, torpoedo fully installed and mated and securing mechanism locked in place with flex hose connected.

6.3.5 Standby condition. From safe condition; remove muzzle covers, and charge breech mechanism.

6.3.6 Ready condition. From standby condition: Rotate Ready Light ON-OFF Actuator Plate to the ON (up) postion. Remotely or manually train launcher to 45° and place control valve operating mechanism in the breechward position (automatic with remote train) and unlock but do not remove padlock securing Emergency Fire latch plate.

6.3.7 Torpedo tube in stowed position. The stowed position of the torpedo tube is defined as having the torpedo barrels parallel to the ship longitudinal axis and with the muzzle facing forward.



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6.3.8 Torpedo tube in firing position. The firing position of the torpedo tube is defined as having the torpedo tube trained 45 degrees outboard from the stowed position.

6.3.9 Standard item. A standard item is a material, part, component, subassembly, or equipment identified or described in military or adopted federal and industry standards.

Custodian:

Navy - OS

Preparing Activity:

Navy - OS

Project No. 1045-N011



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DATA SHEET

QUALITY CONFORMANCE INSPECTION

TESTS AND EXAMINATIONS

FOR

TORPEDO TUBE MARK 32 MOD 14

(LD 632023)

EQUIPMENT STATUS

| CONTRACTOR CERTIFICATION | | GOVERNMENT ACCEPTANCE | |
|---|---|---|---|
| Contractor | | Date of Acceptance | |
| Contract | | Accepted by | |
| Equipment Ser. No. | | Title | |
| Insp. Req. Satisfied | | Agency | |

ORDALT STATUS

| ORDALT NO. | INSTALLED | ORDALT NO. | INSTALLED |
|---|---|---|---|
| | | | |
| | | | |
| | | | |



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DATA SHEET

QUALITY CONFORMANCE INSPECTION

TORPEDO TUBE MARK 32 MOD 14 Equipment Serial No._____

10. SCOPE

10.1 This appendix contains data sheets to record the results of the Quality Conformance tests and examinations of 4.4 of this specification. When completed, with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the Torpedo Tube assembly bearing the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. When the contract or order specifies use of this appendix, the Quality Conformance Inspection tests of 4.4 shall be completed and the results recorded or verified in these data sheets prior to acceptance of the Torpedo Tube Assembly. Unless otherwise noted, the sequence of testing procedures presented herein is suggested only. The procedures may be performed in any order which will permit the most practical and economical use of manpower and equipment.

20.2 Recording of results. Where space is provided in the "Reading" column of the Quality Conformance Test data sheets, the value or specific reading obtained shall be recorded and the initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated, the individual performing the test or inspection shall initial in the space provided in the "Reading" column indicating that the requirements have been met satisfactorily.

| | Reference | Limits | Reading |
|---|---|---|---|
| 30. QUALITY CONFORMANCE INSPECTION TEST AND EXAMINATION DATA | | | |
| 30.1 Test equipment | 4.4.4 | In calibration | _____ |
| 30.2 Test procedure | 4.4.5 | | |
| 30.2.1 4500 psig hydrostatic pressure, breech mechanism | 4.4.5.1 | | |
| Barrel No. 1 | | Verified | _____ |
| Barrel No. 2 | | Verified | _____ |
| Barrel No. 3 | | Verified | _____ |

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| | Reference | Limits | Reading |
|---|---|---|---|
| 30.2.2 Hydrostatic pressure (low pressure system) | 4.4.5.2 | | |
| 225 psig hydrostatic pressure, low pressure pneumatic system | | | |
| Barrel No. 1 | | Verified | _____ |
| Barrel No. 2 | | Verified | _____ |
| Barrel No. 3 | | Verified | _____ |
| 30.2.3 Pneumatic system pressure drop | 4.4.5.3 | | |
| Securing mechanism removed | | | |
| Barrel No. 1 | | 1350 psig min | _____ |
| Barrel No. 2 | | 1350 psig min | _____ |
| Barrel No. 3 | | 1350 psig min | _____ |
| Securing mechanism installed | | | |
| Barrel No. 1 | | 750 psig min | _____ |
| Barrel No. 2 | | 750 psig min | _____ |
| Barrel No. 3 | | 750 psig min | _____ |
| 30.2.4 Breech mechanism air loss | 4.4.5.4 | | |
| Barrel No. 1 | | 80.8 lbs max pressure loss | _____ |
| Barrel No. 2 | | 80.8 lbs max pressure loss | _____ |
| Barrel No. 3 | | 80.8 lbs max pressure loss | _____ |
| 30.2.5 Bore size | 4.4.5.5 | | |
| Barrel No. 1 | | 150 lbs max | _____ |
| Barrel No. 2 | | 150 lbs max | _____ |
| Barrel No. 3 | | 150 lbs max | _____ |
| 30.2.6 Dielectric strength | | | |
| Bulkhead mounted control box assy | 4.4.5.6.1 | No Breakdown | _____ |
| SVTT Mk 32 Mod 14 assy | 4.4.5.6.2 | No Breakdown | _____ |
| 30.2.7 Insulation resistance | 4.4.5.7 | | |
| Safe/Ready circuit | | | |
| TB102-1, 2, 3 and 4 | | 1 megohm min | _____ |
| Remain circuits | | 3 megohm min | _____ |



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| | | Reference | Limits | Reading |
|---|---|---|---|---|
| 30.2.8 | Ground circuit continuity | 4.4.5.8 | Verified | _____ |
| | Continuity between TB102-14 and pin G of J2 | | Verified | _____ |
| | Continuity between junction box 5185872 and control box on: | | | |
| | Barrel No. 1 | | Verified | _____ |
| | Barrel No. 2 | | Verified | _____ |
| | Barrel No. 3 | | Verified | _____ |
| | Continuity between junction box 5206053 and control box on: | | | |
| | Barrel No. 1 | | Verified | _____ |
| | Barrel No. 2 | | Verified | _____ |
| | Barrel No. 3 | | Verified | _____ |
| | Continuity between TB201-16 of Mount Barrel Junction Box and the box. | | Verified | _____ |
| | Continuity between TB304-12 of Bulkhead Mounted Control Box and the box. | | Verified | _____ |
| 30.2.9 | Firing solenoid | 4.4.5.9 | | |
| | Barrel No. 1 | | 1.5 amps max | _____ |
| | Barrel No. 2 | | 1.5 amps max | _____ |
| | Barrel No. 3 | | 1.5 amps max | _____ |
| 30.2.10 | Safe/Ready solenoid | 4.4.5.10 | | |
| | Barrel No. 1 | | 0.2 amps max | _____ |
| | Barrel No. 2 | | 0.2 amps max | _____ |
| | Barrel No. 3 | | 0.2 amps max | _____ |
| 30.2.11 | Squib fire switch | 4.4.5.11 | | |
| | Barrel No. 1 | | Proper operation | _____ |
| | Barrel No. 2 | | Proper operation | _____ |
| | Barrel No. 3 | | Proper operation | _____ |
| 30.2.12 | Interlock switch | 4.4.5.12 | Open at 0° ± 2.5° | _____ |
| | | | Closed at 45° ± 2.5° | _____ |
| 30.2.13 | Training crank | 4.4.5.13 | 30 lbs max | _____ |
| 30.2.14 | Muzzle cover | 4.4.5.14 | | |
| | Barrel No. 1 | | 30 ± 10 lbs | _____ |
| | Barrel No. 2 | | 30 ± 10 lbs | _____ |
| | Barrel No. 3 | | 30 ± 10 lbs | _____ |



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| | Reference | Limits | Reading |
|---|---|---|---|
| 30.2.15 Breakaway plug puller mechanism | 4.4.5.15 | | |
| Barrel No. 1 | | Hand Latchable, Hand Unlatchable | _____ |
| Barrel No. 2 | | Hand Latchable, Hand Unlatchable | _____ |
| Barrel No. 3 | | Hand Latchable, Hand Unlatchable | _____ |
| 30.2.16.1/2 Pressure switch | 4.4.5.16 | | |
| Barrel No. 1 switch closure | | 1275 ± 75 psig | _____ |
| Barrel No. 1 switch opening | | 1200 psig min | _____ |
| Barrel No. 2 switch closure | | 1275 ± 75 psig | _____ |
| Barrel No. 2 switch opening | | 1200 psig min | _____ |
| Barrel No. 3 switch closure | | 1275 ± 75 psig | _____ |
| Barrel No. 3 switch opening | | 1200 psig min | _____ |
| 30.2.16.3 Ready light switch | 4.4.5.16 | | |
| Barrel No. 1 | | Switch opens | _____ |
| Barrel No. 2 | | Switch opens | _____ |
| Barrel No. 3 | | Switch opens | _____ |
| 30.2.17 Dry firing test | 4.4.5.17 | Breech mechanisms function in all barrels | _____ |
| Electrical firing | 4.4.5.17.1 | | |
| Barrel No. 1 | | One 400-Millisecond voltage application per firing | _____ |
| Barrel No. 2 | | One 400-Millisecond voltage application per firing | _____ |
| Barrel No. 3 | | One 400-Millisecond voltage application per firing | _____ |
| Manual firing | 4.4.5.17.2 | | |
| Barrel No. 1 | | 25 lbs max | _____ |
| Barrel No. 2 | | 25 lbs max | _____ |
| Barrel No. 3 | | 25 lbs max | _____ |



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| | Reference | Limits | Reading |
|---|---|---|---|
| Breakaway plug mechanism | 4.4.5.17.3 | | |
| Barrel No. 1 | | Latched, hand Unlatchable | _____ |
| Barrel No. 2 | | Latched, hand Unlatchable | _____ |
| Barrel No. 3 | | Latched, hand Unlatchable | _____ |
| Control valve operating lever | 4.4.5.17.4 | | |
| Barrel No. 1 | | No leakage, will not fire | _____ |
| Barrel No. 2 | | No leakage, will not fire | _____ |
| Barrel No. 3 | | No leakage, will not fire | _____ |

| | Reference | Limits | Reading |
|---|---|---|---|
| Pressure switch operation | 4.4.5.17.5 | | |
| Barrel No. 1 | | Operation per Schedule I | _____ |
| Barrel No. 2 | | Operation per Schedule I | _____ |
| Barrel No. 3 | | Operation per Schedule I | _____ |
| Breakaway plug mechanism | 4.4.5.17.6 | | |
| Barrel No. 1 | | Permits firing, remains latched | _____ |
| Barrel No. 2 | | Permits firing, remains latched | _____ |
| Barrel No. 3 | | Permits firing, remains latched | _____ |
| Observation for damage | 4.4.5.17.7 | | |
| Barrel No. 1 | | No damage | _____ |
| Barrel No. 2 | | No damage | _____ |
| Barrel No. 3 | | No damage | _____ |
| 30.2.18 Continuity, systems inspection tests | 4.4.5.18 | Continuity per schematic | _____ |



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| | | Reference | Limits | Reading |
|---|---|---|---|---|
| 30.2.18.1 | Bulkhead Mounted Control Box (BMCB) | | | |
| | (1) Set ohmmeter to R x 1 scale and verify continuity at 24 terminals (A thru Z excluding i and o) between J301 and J304 connectors. | | 1 ohm max | _____ |
| | (2) Supply 26 Vdc between T7 on TB 301 and T7 on TB 303. With relay K301 energized, verify continuity* between: | | | |
| | (a) terminals 1 thru 4 on TB301 and terminals 1 thru 4 on TB303. | | 1 ohm max | _____ |
| | (b) T5 and T6 on TB301. | | 1 ohm max | _____ |
| | (c) T5 and T6 on TB303. | | 1 ohm max | _____ |
| | *With ohmmeter set to R x 1 scale. | | | |
| | (3) With Panel Control Switch in LOCAL position,* verify continuity** between: | | | |
| | (a) terminals A on J303 and R on J305 | | 1 ohm max | _____ |
| | (b) T2 and T4 on TB304. | | 1 ohm max | _____ |
| | (c) T11 and T3 on TB304. | | 1 ohm max | _____ |
| | (d) With firing door control switch in OPEN DOOR position T3 on TB302 and T9 on TB304. | | 1 ohm max | _____ |
| | *Tube train control switch in TRAIN OUT position, and firing door control switch in CLOSE DOOR position. **With ohmmeter set to Rx1 scale | | | |
| | (4) Supply 115 Vac 60 Hz, 10 between terminals 4 and 5 on TB304 and verify lamp operation on each position of the 3 switches. | | Verify | _____ |



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| | Reference | Limits | Reading |
|---|---|---|---|
| (5) With Panel Control Switch in REMOTE position, supply 115VAC 60 Hz, 10 to: | | | |
| (a) Terminals 6 & 7 on TB 302 and verify continuity (with ohmmeter set to R x 1 scale) between T 3 on TB 302 and T 9 on TB 304. | | 1 ohm max | _____ |
| (b) Terminals 4 & 5 on TB 302 and verify continuity (with ohmmeter set to R X 1 scale) between T 3 on TB 302 and T 2 on TB 304. | | 1 ohm max | _____ |
| (c) Terminals 11 & 12 on TB 303 and verify that Relay K 303 is open by infinity reading on ohmmeter between Terminals 8 & 9 on TB 302. | | 1 ohm max | _____ |
| 30.2.18.2 Mount Barrel Junction Box (MBJB) | | | |
| (1) Supply 115 V, 60 Hz, 10 to Terminals 10 & 11 on TB 202 and verify continuity (with ohmmeter set to R X 1 scale) between: | | | |
| (a) TE on J 202 and TB on J204. | | 1 ohm max | _____ |
| (b) TD on J 202 and TC on J204. | | 1 ohm max | _____ |
| (2) Supply 115 V, 60 Hz 10 to Terminals 11 & 12 on TB 202 and verify continuity (with ohmmeter set to R X 1 scale) between: | | | |
| (a) TD on J 202 and TB on J 204. | | 1 ohm max | _____ |
| (b) TE on J 202 and TC on J 204. | | 1 ohm max | _____ |



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| | Reference | Limits | Reading |
|---|---|---|---|
| (3) With ohmmeter set to R x 1 scale, verify continuity between the following terminals: | | | |
| TA on J 201 and TA on J 203 | | 1 ohm max | _____ |
| TE on J 201 and TA on J 206 | | 1 ohm max | _____ |
| TB on J 201 and TA on J 207 | | 1 ohm max | _____ |
| TF on J 201 and TB on J 203 | | 1 ohm max | _____ |
| TF on J 201 and TB on J 206 | | 1 ohm max | _____ |
| TF on J 201 and TB on J 207 | | 1 ohm max | _____ |
| TG on J 201 and T 13 on TB 201 | | 1 ohm max | _____ |
| TC on J 201 and T 14 on TB 201 | | 1 ohm max | _____ |
| TH on J 201 and TA on J 205 | | 1 ohm max | _____ |
| TD on J 201 and TA on J 208 | | 1 ohm max | _____ |
| TV on J 201 and TA on J 202 | | 1 ohm max | _____ |
| TQ on J 201 and TH on 202 | | 1 ohm max | _____ |
| TR on J 201 and TE on J 205 | | 1 ohm max | _____ |
| TZ on J 201 and T16 on TB 201 to GR | | 1 ohm max | _____ |
| TS on J 201 and TI on J 205 | | 1 ohm max | _____ |
| TX on J 201 and TJ on J 205 | | 1 ohm max | _____ |
| T 11 on TB 201 and T 10 on TB 202 | | 1 ohm max | _____ |
| T 12 on TB 202 and TH on J 205 | | 1 ohm max | _____ |

30.2.18.3 BMCB/SVTT inspection tests. Secure tube assembly test stand.

(1) Visually verify existence of local-remote panel control switch on the BMCB Mk 60 Mod 0

(2) Local operation - with the local-remote switch in LOCAL position:

(a) Supply 115 V, 60 Hz, 10 and 115 V, 60 Z, 30 power to base mount assy. thru cable MIL-C-915/8 types SS-Z.

(b) Connect SA 2883180 (test cabinet/power supply) via plugs P 302 and P 303 to BMCB. Turn test cabinet power switch on.

(c) Place fire door control switch on BMCB to OPEN DOOR position. Verify OPEN DOOR lamp lit in test cabinet. Verify _____



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| | Reference | Limits | Reading |
|---|---|---|---|
| (d) Close DOOR OPEN switch in test cabinet. Turn TUBE TRAIN CONTROL switch to TRAIN and verify train out of tube to firing position. | | Verify | _____ |
| (e) With tube in firing position: | | | |
| 1. Close inter-lock supply common switch in test cabinet to initiate readying sequence. Close S 203 Limit switch manually. Verify: | | | |
| a. Operation of safe/ready solenoid valves on all 3 barrels. | | Verify | _____ |
| b. Firing sector clear lamp lit in test cabinet. | | Verify | _____ |

2. Supply 115 V 60 Hz, 10 power to terminals listed and pressurize each Tube Firing circuit 1200 PSI min. Verify operation of main firing solenoids.

| Supply Power to Junction Box & Frame Assy. | Solenoids in Control Box | | |
|---|---|---|---|
| TB 1<br>TB 10 & 11 | No 1 | Verify | _____ |
| TB 2<br>TB 10 & 11 | No 2 | Verify | _____ |
| TB 3<br>TB 10 & 11 | No 3 | Verify | _____ |

(f) Remove power from junction box and frame assy.; open INTER-LOCK SUPPLY COMMON switch in test cabinet and turn TUBE TRAIN CONTROL switch to STOW position. With tube in stowed position:

1. Open DOOR OPEN switch in test cabinet. Turn FIRING DOOR CONTROL switch to CLOSE DOOR position and verify that the CLOSE DOOR lamp in test cabinet is lit. Verify _____



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| | Reference | Limits | Reading |
|---|---|---|---|
| 2. Close PREPARE MOUNT ORDER switch in test cabinet and verify that the OPEN DOOR lamp in test cabinet is not lit. Open switch. | | Verify | _____ |
| (3) Remote Operation-With the LOCAL-REMOTE switch in the REMOTE position: | | | |
| (a) | | | |
| 1. Repeat directions per step 4 B (1) & (2). | | | |
| 2. Pressurize tubing as in step 1. | | | |
| 3. Place all 3 ready switch actuator plates to ON. | | | |
| (b) Close BMCB IN REMOTE SUPPLY switch in test cabinet and verify BMCB IN REMOTE lamp in test cabinet is lit. | | Verify | _____ |
| (c) Close PREPARE MOUNT ORDER switch in test cabinet and verify OPEN DOOR lamp in test cabinet is lit. | | Verify | _____ |
| (d) Close DOOR OPEN switch in test cabinet to train out tube to firing position. | | | |
| (e) Remote readying - Close the INTER-LOCK SUPPLY COMMON switch in test cabinet and verify execution of the Remote Ready sequence, i.e.: | | | |
| 1. Actuation of all 3 safe/ ready solenoids. | | Verify | _____ |
| 2. Actuation of all 3 safe/ ready cylinders. | | Verify | _____ |
| 3. Actuation of all 3 control valve operating levers fully breechward to the ready position. | | Verify | _____ |
| | | Verify | _____ |



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| | Reference | Limits | Reading |
|---|---|---|---|
| 4. Continuity between terminals 5 & 6 on TB 105 on barrel #1, 1 & 2 on TB 104 on barrel #2, 9 & 10 on TB 104 on Barrel #3. Note*. 1 ohm max. | | Verify | _____ |

*NOTE: Firing Circuit must be pressurized sufficiently to close each pressure inter-lock switch. 1200 PSI min.

(f) In the test cabinet:

1. Open INTER-LOCK SUPPLY COMMON switch.

2. Open DOOR OPEN switch.

3. Open BMCB IN REMOTE SUPPLY switch.

4. Open PREPARE MOUNT ORDER switch.

5. Close STOW MOUNT ORDER switch to return tube to stow position. Verify CLOSE DOOR lamp in test cabinet is lit. Verify _____

6. Open STOW MOUNT ORDER switch and release all air pressure.

(4) Power Azimuth Train -

(a) With the LOCAL-REMOTE switch on BMCB in LOCAL position and the DOOR OPEN switch in test cabinet closed, turn the TUBE TRAIN CONTROL switch on BMCB to TRAIN. The tube shall train through a 45 degree arc from the stowed position to the fire position in 15±10 seconds.

Actual Train Time_____ Verify _____



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| | Reference | Limits | Reading |
|---|---|---|---|

(b) Airborne Noise Levels - Place microphone of sound level meter/ octave analyzer approximately 1' above center line of tube train at test stand. Electrically train and stow tube at a time when the background noise is at a minimum. (Suggest 3rd shift.) During normal operation, the sound pressure level of airborne noise shall not exceed the following levels at the center frequency listed by more than two decibels.

| Center Frequency of Octave Bands. Hertz | 32 | 63 | 125 | 250 | 500 | 1000 | 2000 | 4000 | 8000 |
|---|---|---|---|---|---|---|---|---|---|
| Airborne Noise Level in db. (0.0002 dynes/$CM^2$) | 105 | 100 | 95 | 90 | 90 | 85 | 85 | 85 | 85 |
| Actual Noise Level | | | | | | | | | |

30.2.19 Lanyard retainers 4.4.5.19

| | | |
|---|---|---|
| Barrel No. 1 | Hand depressable, secures wire loop | _____ |
| Barrel No. 2 | Hand depressable, secures wire loop | _____ |
| Barrel No. 3 | Hand depressable, secures wire loop | _____ |

30.2.20 Torpedo exit velocity 4.4.5.20
(3) barrels

Barrel No. 1

| Launchings per Barrel (No.) | Torpedo Type (Mk) | (Mod) | Air Flask Pressure (psig) | Pressure Switch Condition | Firing Method | Min. Exit Velocity (Ft/sec) | Reading Obtained |
|---|---|---|---|---|---|---|---|
| 1 | 46 | 1 | 1200 | - | Manual | 40 | _____ |
| 1 | 46 | 1 | 1350 | Closed | Electrical | 40 | _____ |
| 1 | 46 | 1 | 1500 | Closed | Electrical | 40 | _____ |



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Barrel No. 2 4.4.5.20

| Launchings Per Barrel (No.) | Torpedo Type (Mk) | (Mod) | Air Flask Pressure (psig) | Pressure Switch Condition | Firing Method | Min. Exit Velocity (ft/sec) | Reading Obtained |
|---|---|---|---|---|---|---|---|
| 1 | 46 | 1 | 1200 | - | Manual | 40 | _____ |
| 1 | 46 | 1 | 1350 | Closed | Electrical | 40 | _____ |
| 1 | 46 | 1 | 1500 | Closed | Electrical | 40 | _____ |

Reference

Barrel No. 3 4.4.5.20

| Launchings Per Barrel (No.) | Torpedo Type (Mk) | (Mod) | Air Flask Pressure (psig) | Pressure Switch Condition | Firing Method | Min. Exit Velocity (ft/sec) | Reading Obtained |
|---|---|---|---|---|---|---|---|
| 1 | 46 | 1 | 1200 | - | Manual | 40 | _____ |
| 1 | 46 | 1 | 1350 | Closed | Electrical | 40 | _____ |
| 1 | 46 | 1 | 1500 | Closed | Electrical | 40 | _____ |

| | Reference | Limits | Reading |
|---|---|---|---|
| Electrical firing | 4.4.5.21 | | |
| Barrel No. 1 | | One 400-millisecond voltage application per firing | _____ |
| Barrel No. 2 | | One 400-millisecond voltage application per firing | _____ |
| Barrel No. 3 | | One 400-millisecond voltage application per firing | _____ |



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| | Reference | Limits | Reading |
|---|---|---|---|
| Manual firing | 4.4.5.22 | | |
| Barrel No. 1 | | 25 lbs max | _____ |
| Barrel No. 2 | | 25 lbs max | _____ |
| Barrel No. 3 | | 25 lbs max | _____ |
| Breakaway plug mechanism | 4.4.5.23 | | |
| Breakaway plug mechanism | 4.4.5.23.1 | | |
| Barrel No. 1 | | Unlatched, attached to torpedo receptacle | _____ |
| Barrel No. 2 | | Unlatched, attached to torpedo receptacle | _____ |
| Barrel No. 3 | | Unlatched, attached to torpedo receptacle | _____ |
| Breakaway plug | 4.4.5.23.2 | | |
| Barrel No. 1 | | Latched, hand Unlatchable | _____ |
| Barrel No. 2 | | Latched, hand Unlatchable | _____ |
| Barrel No. 3 | | Latched, hand Unlatchable | _____ |
| Breakaway plug | 4.4.5.23.3 | | |
| Barrel No. 1 | | No damage after each firing | _____ |
| Barrel No. 2 | | No damage after each firing | _____ |
| Barrel No. 3 | | No damage after each firing | _____ |
| Pressure switch operation | 4.4.5.24 | | |
| Barrel No. 1 | | Operation per Schedule II | _____ |
| Barrel No. 2 | | Operation per Schedule II | _____ |
| Barrel No. 3 | | Operation per Schedule II | _____ |
| Control valve operating lever | 4.4.5.25 | | |
| Barrel No. 1 | | No leakage | _____ |
| Barrel No. 2 | | No leakage | _____ |
| Barrel No. 3 | | No leakage | _____ |



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| | | | |
|---|---|---|---|
| Observation for damage | 4.4.5.26 | | |
| Barrel No. 1 | | No damage | _____ |
| Barrel No. 2 | | No damage | _____ |
| Barrel No. 3 | | No damage | _____ |

30.3 Packaging, packing and marking

| | | |
|---|---|---|
| 30.3.1 | Preservation and packaging | 5.1 |
| 30.3.2 | Packing | 5.2 |
| 30.3.3 | Marking | 5.3 |



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-T-82726(OS) | TORPEDO TUBE, SURFACE VESSEL, MARK 32 MOD 14 |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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